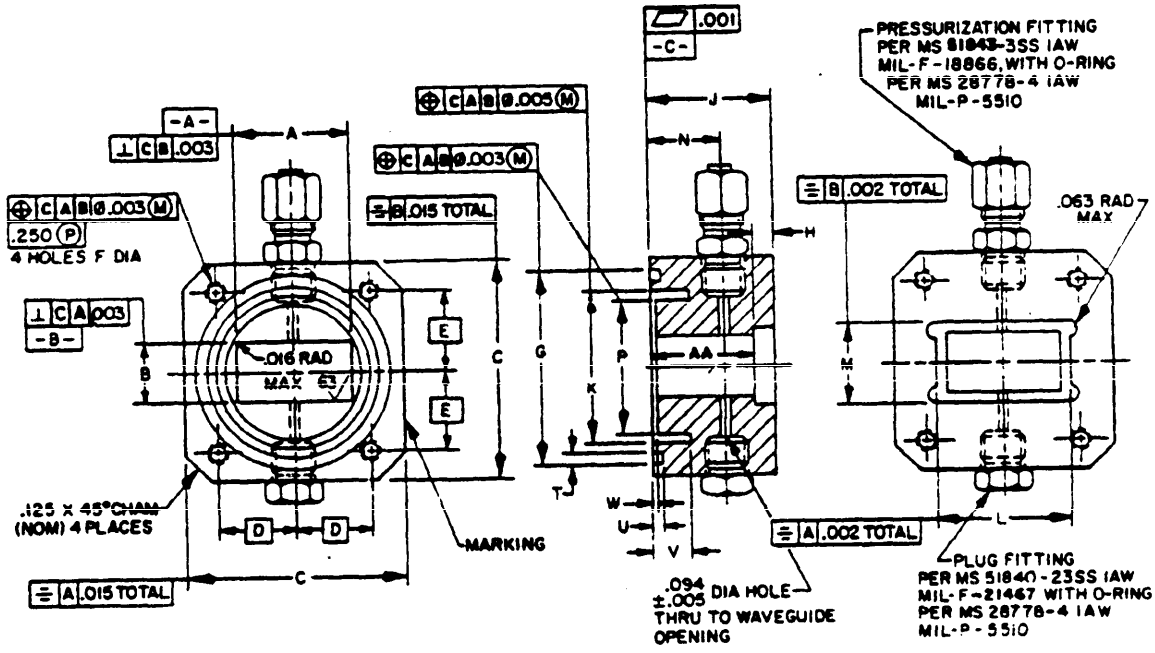
MILITARY SPECIFICATION SHEET

FLANGES, WAVEGUIDE (CHOKE) (WITH PRESSURIZATION FITTING)

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the flanges described herein shall consist of this document and the latest issue of Specification MIL-F-3922.

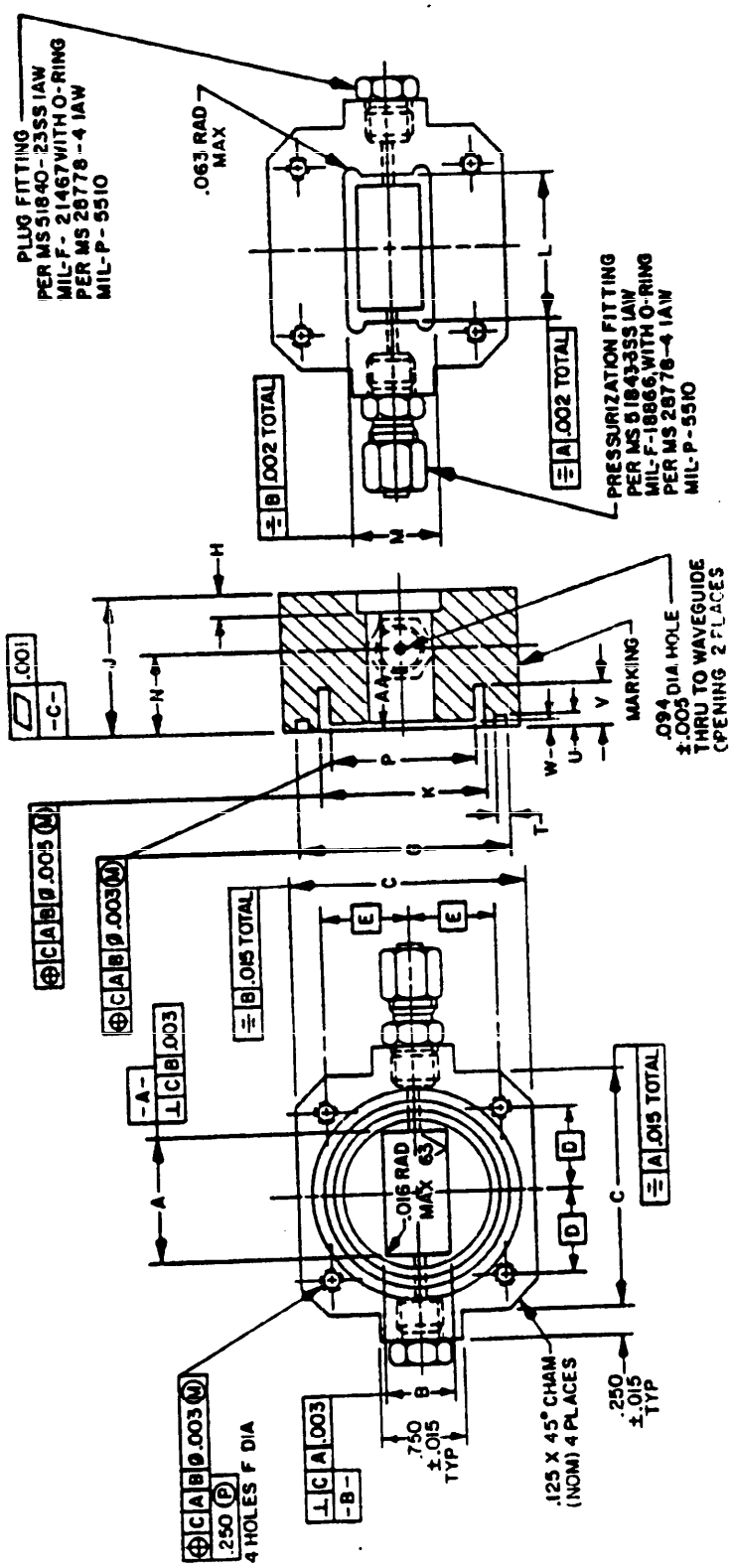


Configuration 1. Socket mount flange with fittings in broad wall of waveguide opening (part numbers -01 and -02).

FIGURE 1. Flanges.

FSC 5985

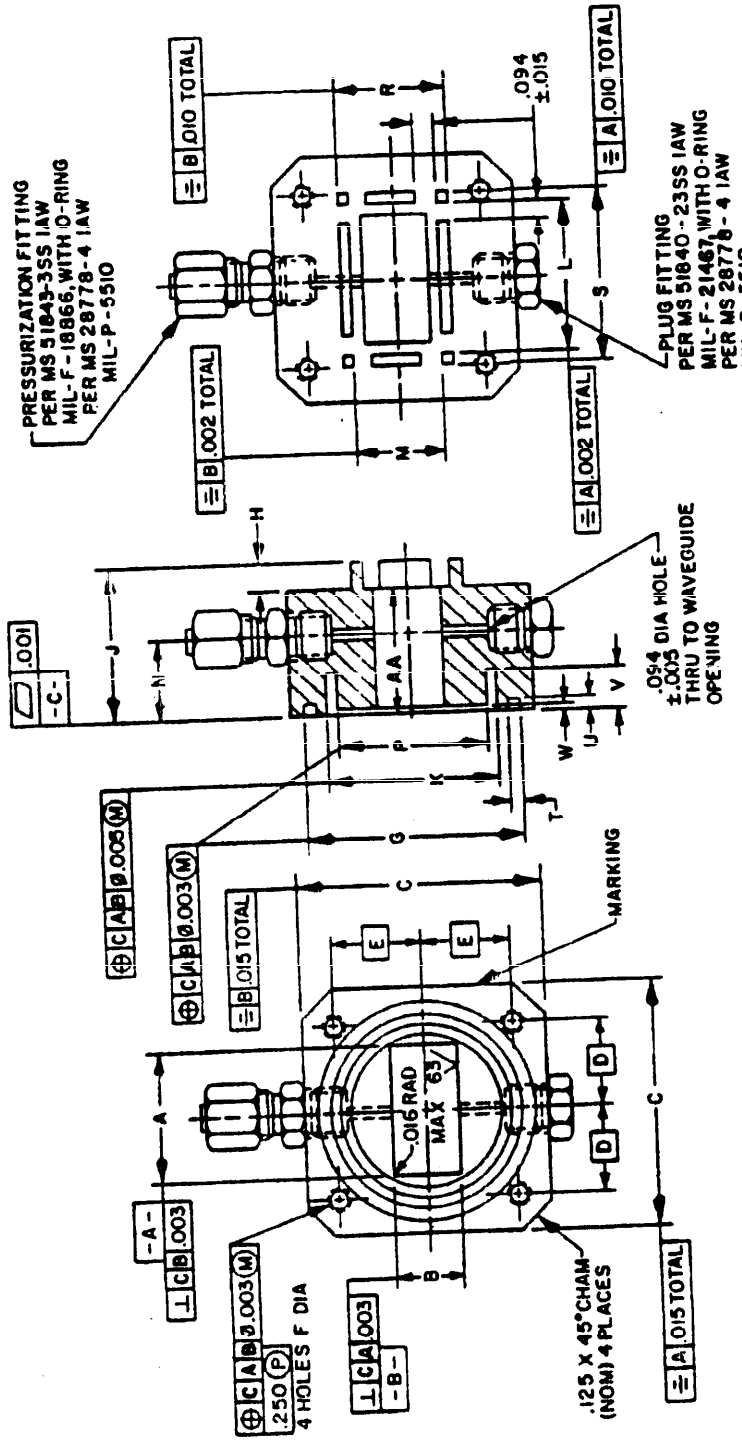
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Configuration 2. Socket mount flange with fittings in narrow wall of waveguide opening (part numbers -03 and -04).

FIGURE 1. Flanges - Continued.

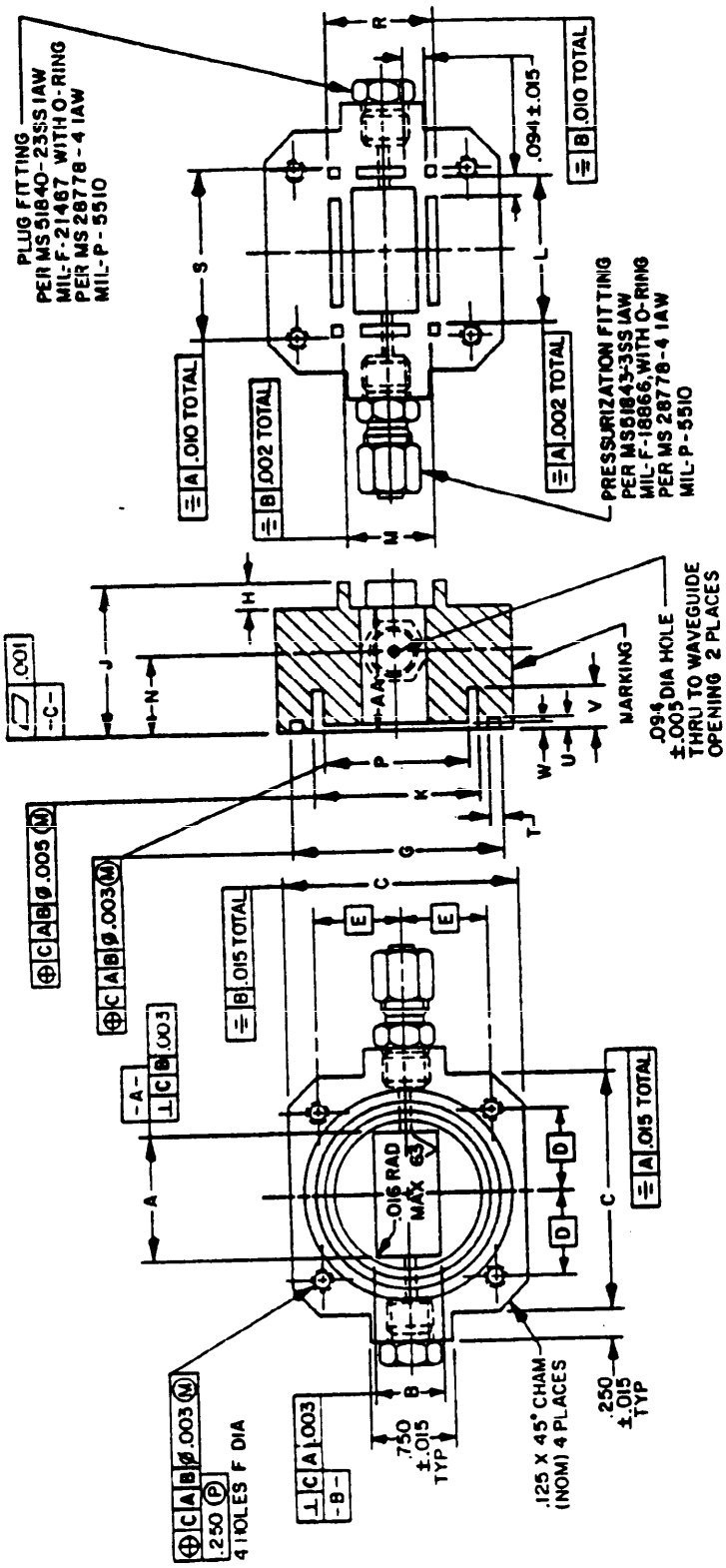
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Configuration 3. Butt mount flange with fittings in broad wall of waveguide opening (part numbers -05 and -06).

FIGURE 1. Flanges - Continued.

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Configuration 4. Butt mount flange with fittings in narrow wall of waveguide opening (part numbers -07 and -08).

FIGURE 1. Flanges - Continued.

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TABLE I. Part number characteristics and dimensions.

Part number M3922/73			Part number M3922/73			Config-uration No.	Dimensions							
	Used with			Used with										
Dash No. 1/	Waveguide M85/1-	Mating flange M3922/70-	Dash No. 2/	Waveguide M85/1-	Mating flange M3922/70-		A ±.003 (.08)	B ±.003 (.08)	C ±.015 (.38)	D BSC.	E BSC.	F	G Dia. ±.003 (.08)	H ±.015 (.38)
01	155, 156	001, 013	02	157, 158, 160	002, 014	1	1.020 (25.91)	.510 (12.95)	1.687 (42.85)	.640 (16.26)	.670 (17.02)	.164-32 UNC-2B	1.634 (41.50)	.125 (3.18)
03	155, 156	001, 013	04	157, 158, 160	002, 014	2	1.020 (25.91)	.510 (12.95)	1.687 (42.85)	.640 (16.26)	.670 (17.02)	.164-32 UNC-2B	1.634 (41.50)	.125 (3.18)
05	155, 156	001, 013	06	157, 158, 160	002, 014	3	1.020 (25.91)	.510 (12.95)	1.687 (42.85)	.640 (16.26)	.670 (17.02)	.164-32 UNC-2B	1.634 (41.50)	.250 (6.35)
07	155, 156	001, 013	08	157, 158, 160	002, 014	4	1.020 (25.91)	.510 (12.95)	1.687 (42.85)	.640 (16.26)	.670 (17.02)	.164-32 UNC-2B	1.634 (41.50)	.250 (6.35)

Dash No. 1/	Dash No. 2/	Dimensions												
		J ±.015 (.38)	K Dia. ±.005 (.13)	L	M	N ±.005 (.13)	P	R ±.010 (.25)	S ±.010 (.25)	T ±.003 (.08)	U ±.002 (.05)	V ±.003 (.08)	W ±.0010 (.03)	AA ±.005 (.13)
01	02	1.000 (25.40)	1.358 (34.49)	1.153 +.003 -.000 (29.29 (.08) -.000)	.643 +.003 -.000 (16.33 (.08) -.000)	.591 (15.01)	1.165 +.005 -.000 (29.59 (.13) -.000)	---	---	.113 (2.87)	.065 (1.65)	.312 (7.92)	.0125 (.32)	.875 (22.22)
03	04	1.000 (25.40)	1.358 (34.49)	1.153 +.003 -.000 (29.29 (.08) -.000)	.643 +.003 -.000 (16.33 (.08) -.000)	.591 (15.01)	1.165 +.005 -.000 (29.59 (.13) -.000)	---	---	.113 (2.87)	.065 (1.65)	.312 (7.92)	.0125 (.32)	.875 (22.22)
05	06	1.125 (28.58)	1.358 (34.49)	1.153 +.003 -.000 (29.29 (.08) -.000)	.643 +.003 -.000 (16.33 (.08) -.000)	.591 (15.01)	1.165 +.005 -.000 (29.59 (.13) -.000)	.891 (22.63)	1.431 (36.35)	.113 (2.87)	.065 (1.65)	.312 (7.92)	.0125 (.32)	.875 (22.22)
07	08	1.125 (28.58)	1.358 (34.49)	1.153 +.003 -.000 (29.29 (.08) -.000)	.643 +.003 -.000 (16.33 (.08) -.000)	.591 (15.01)	1.165 +.005 -.000 (29.59 (.13) -.000)	.891 (22.63)	1.431 (36.35)	.113 (2.87)	.065 (1.65)	.312 (7.92)	.0125 (.32)	.875 (22.22)

1/ Material shall be copper alloy.
2/ Material shall be aluminum alloy.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. Roughness of mating surfaces shall not exceed 63 microinches in accordance with ANSI-B46.1-1962, except that flaws shall be included in the roughness height measurement.

INCHES	MM	INCHES	MM
.001	.03	.016	.41
.002	.05	.063	1.60
.003	.08	.094	2.39
.005	.13	.125	3.18
.010	.25	.250	6.35
.015	.38	.750	19.05

FIGURE 1. Flanges - Continued.

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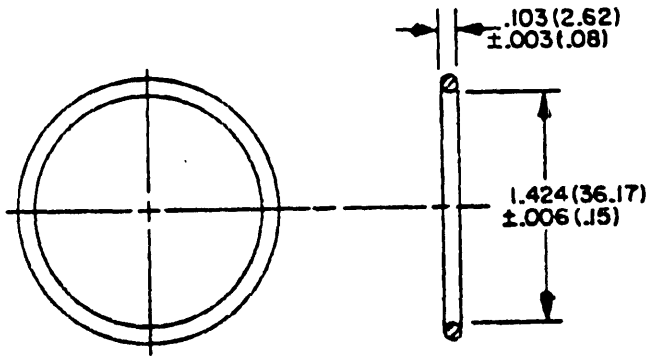


FIGURE 2. Gasket.

TABLE II. Items supplied with flanges.

Item	Items supplied with part number		Quantity
	M3922/ -01, 02, 03, 04	M3922/ -05, 06, 07, 08	
Gasket	See figure 2	See figure 2	1
Socket head cap screw	.164-32 UNC-2A 1.5 inch long	.164-32 UNC-2A 1.75 inch long	4
Spring lockwasher	.164 x .040 thick	.164 x .040 thick	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.

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REQUIREMENTS:

Dimensions and configuration:
Flange: See configurations 1 through 4 of flange 1 and table I.
Gasket: See figure 2 and table II.

Material:
Flange: Metallic alloy as specified in table I.
Socket head cap screw and spring lockwasher: Corrosion resisting steel in accordance with QQ-S-763, 300 series.

Surface roughness: The surface roughness for the four inside walls of each flange shall be the same as the surface roughness requirements of the interfacing waveguide.

Marking: See configurations 1 through 4.

Materials supplied with flanges: See table II.

Pressurization: Flanges shall be capable of holding a pressure of 30 lb_f/in^2 gage of dry nitrogen gas for a period of 5 minutes without leakage.

Note: The waveguide flanges listed in this specification are to be supplied with the associated pressurization fittings installed in the flanges.

Part number: M3922/73- (dash number from table I)

ENGINEERING INFORMATION:

Frequency range: See table III.

TABLE III. Engineering information.

Part number M3922/73-	Frequency range (GHz)
01,02,03,04, 05,06,07,08	7.0-11.0

Supersession data: See table IV.

TABLE IV. Supersession data.

Superseded type	Superseding part number M3922/73-
UG-1911/U	05

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-N442-3)

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